



Insurance on the Cloud

How AWS is helping insurers address some of their top business priorities





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Foreword

The insurance industry is navigating a turbulent landscape, as new buying patterns, evolving customer expectations, inflation and interest rate pressures, and rising claims costs require insurers to have higher levels of agility and insight.

In response, insurers are focused on modernizing their mission-critical applications and infrastructure and driving innovation in their organizations through technology. And, they're increasingly looking to the cloud to power their technology and business transformation.

As a pioneer at the intersection of financial services and cloud technology, Amazon Web Services (AWS) is helping insurance organizations—from the largest global insurers and reinsurers to the fastest-growing InsurTechs—reinvent the future of insurance.

Leveraging AWS services and solutions from our global community of partners, insurers are becoming more efficient, increasing speed to market for new products, and harnessing data to deliver highly-personalized customer experiences, as well as make more informed business decisions.

We wrote this ebook to share actionable insights based on our work with customers and partners in the insurance industry. We provide perspective on five critical business priorities for insurers and explore how they're using the cloud to continue to grow and thrive in a rapidly-changing landscape.

We welcome your feedback and questions and want to know where we can dive deeper to help you on your own transformation journey.

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INTRODUCTION

Forces driving the need for broader and faster digital-led transformation

The insurance industry has traditionally evolved cautiously and with intention. However, insurers today face a confluence of external forces that require broader and faster digital-led transformation.

Regulatory oversight continues to expand, increasing compliance complexity and costs.

Customers expect more, as other industries offer the ability to engage easily and immediately in any channel.

Competitive pricing pressures are escalating, especially in personal lines, straining underwriting performance and combined ratios.

Claims frequency and severity are growing due to changing driving patterns, more expensive vehicles and property, and severe weather events. Rising inflation further escalates the cost of claims.

Innovation from InsurTechs, as well as established players, is giving consumers more choice. New models, like usage-based and embedded insurance, are also disrupting traditional paradigms.

The “Great Retirement” is rapidly approaching, so insurance firms must find ways to attract younger and tech-savvy talent in a highly-competitive environment.

In the face of these external factors, historical risk models are revealing their limitations. Insurers need to access broader data sets, including more external and unstructured data, to enrich their historical data and improve their ability to identify and price risk into the future.

Powering transformation with cloud

Given the current environment, insurers are no longer evaluating if they should move to the cloud, but rather how to start, accelerate, or scale their cloud adoption. Insurers are leveraging the cloud to reduce the burden of their legacy estates, primarily for cost savings and increased agility, and to drive innovation and business growth by harnessing data and developing new digital products, experiences, and distribution channels.

This ebook describes how AWS is working with insurers to lay the technology foundation they need to achieve some of these priorities.



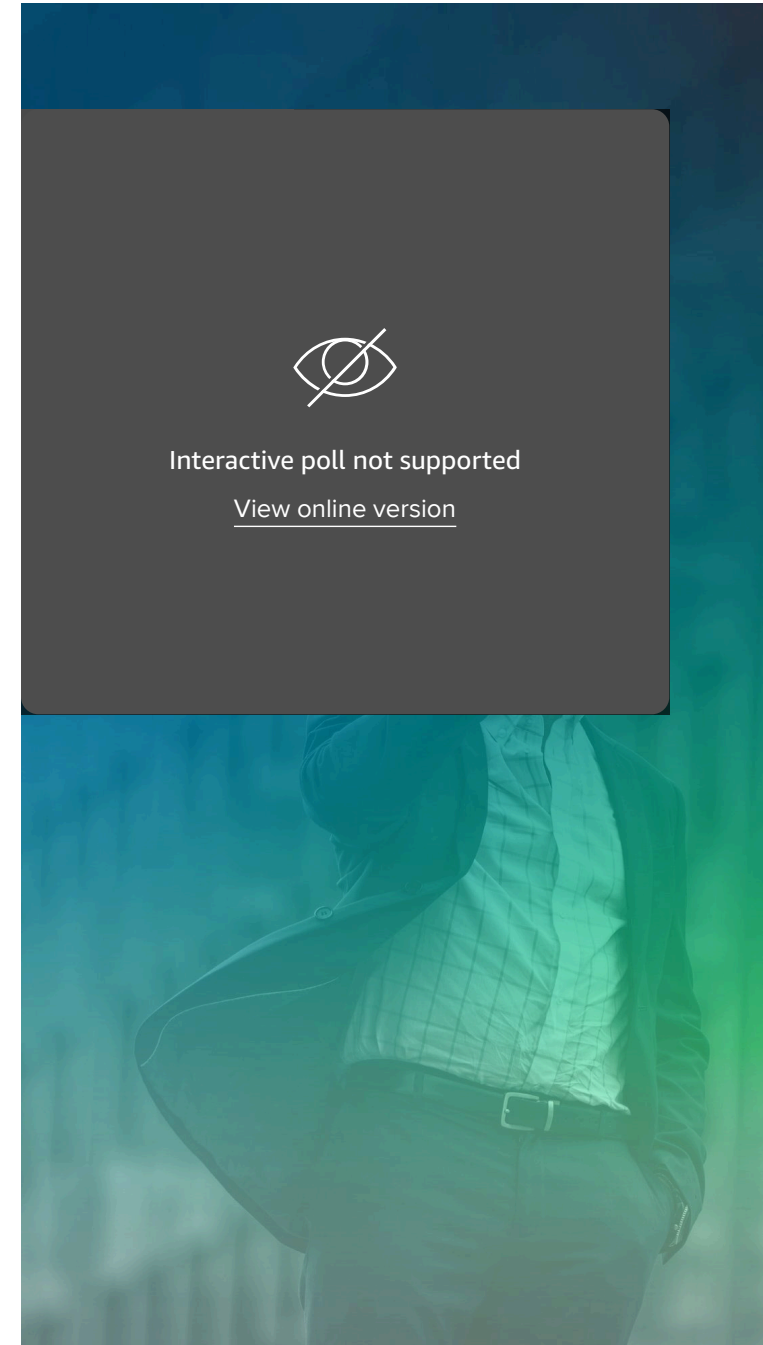
Insurers today face a confluence of external forces that require broader and faster digital-led transformation.”

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PRIORITY 1

Unleash Product Innovation

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PRIORITY 1

Unleash product innovation

Insurers are expanding their product portfolios to fit changing customer demographics and digital purchasing patterns. They, therefore, need their core systems to be agile and flexible enough to develop, test, and scale innovative new products—from cyber and parametric insurance to simple life and annuity products that can be purchased online. Additionally, they need to capture more attributes, either directly from customers or indirectly from third-party data, to drive better rating, as well as feed internal risk selection and portfolio management functions.

Legacy core insurance systems for underwriting, policy, billing, claims administration, document management, and customer communications were built as monolithic solutions, making them slow and difficult to adapt to these new imperatives. Insurers are migrating to more flexible, scalable microservices-based core platforms in the cloud, delivered by AWS partners, that can deliver the speed and agility they require.

Microservices-based architecture delivers several important benefits, including speeding up development and deployment cycles, improving application maintenance and scalability, increasing resilience, and providing teams with the freedom to work independently, while at the same time collaborating and leveraging reusable code. Cloud-native core systems are built with microservices and offer rapid customization features that can be easily controlled and configured by different business teams to address quick-changing requirements. This enables insurers to take new products to market or update existing products rapidly, based on evolving customer needs, risk appetite, and economic factors.

Benefits of a microservices architecture



RESILIENCE

Service independence increases an application's resistance to failure.



AGILITY

Small, independent teams take ownership of their services, shortening development cycle times.



EASY DEPLOYMENT

Continuous integration and delivery make updates quicker.



REUSABLE CODE

Dividing software into small, well-defined modules enables teams to use functions for multiple purposes.



FLEXIBLE SCALING

Each service can be independently scaled to meet demand for the application feature it supports.



TECHNOLOGICAL FREEDOM

Freedom from a "one-size-fits-all" approach means teams can choose the best tools for each job.

Drivers of core modernization

48%

of insurance IT spending goes toward maintaining legacy systems.¹

70%

of IT team time is spent on projects supporting core applications.²

AWS supports your journey

AWS cloud core solutions are delivered through a broad set of independent software vendor (ISV) partners, supporting market-ready solutions for all lines of business. These partner solutions are all powered by AWS cloud infrastructure and leverage Amazon services, ranging from **Amazon Kendra** to **Amazon SageMaker** to **Amazon Alexa**.

AWS supports insurers in moving core systems from on-premises environments to AWS cloud environments through multiple tactics, whether it's relocating core applications from a vendor to Containers and VMware on AWS, rehosting the vendor's solution on AWS ("lift & shift"), or repurchasing a new core solution to replace an existing vendor solution.

Depending on the insurer's needs, these solutions can be deployed in one of three ways:

1. As a SaaS solution with a fully managed deployment
2. As a managed service offering from an ISV or systems integrator that hosts the applications on AWS for the customer
3. As a self-managed solution where the customer hosts and manages all environments on AWS.

[Explore AWS Core Insurance Partners >](#)

Innovation in action



Guardian Life accelerates product and claims innovation

Guardian Life, known for its innovative use of cloud technologies, looked to accelerate time to market for new products and features, while reducing cost of ownership. It selected Accenture Life Insurance & Annuity Platform (ALIP), a SaaS-based, cloud-enabled life insurance policy administration platform, deployed on AWS. ALIP's cloud-first design leverages containers and microservices for optimal flexibility, scalability, and cloud performance. The solution, running on AWS, also enables Guardian to make changes quickly, such as how a claim is executed, in just hours with no coding needed.

[Learn more >](#)



PRIORITY 2

Transform business intelligence

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PRIORITY 2

Transform business intelligence

Although insurance companies rely on data to make decisions across the value chain, the data has traditionally been siloed and lacked consistent formats. Actuarial, underwriting, claims, customer relationship management, billing, and policy data are all housed in separate legacy systems and different systems are used across product lines and channels. Insurers also have unstructured data, from sources like social media and chatlog transcriptions, that are also housed in separate systems. This fragmentation leads to an inconsistent business reporting landscape and hinders the ability to derive business value from the data. Even more significant is a lack of a holistic view of the customer journey.

There is also the opportunity for insurers to take advantage of the universe of data from connected devices, like smartphones, cars, home assistants, wearables, industrial machines, and drones, which can be used to measure risk and prevent losses. Insurance organizations need to have strategies to harness the vast amounts of data available to them from a wide array of sources.

“Insurance organizations need to have strategies to harness the vast amounts of data available to them from a wide array of sources.”

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AWS supports your journey

One of the first steps insurance companies can take to transform business intelligence is to create an AWS data lake to liberate their data from different sources.

By building a data lake on AWS, insurance organizations can integrate disparate internal and external data sources into a consistent data set that is readily available to be categorized and processed, as well as stored cost-effectively, at any scale. AWS data lakes provide a foundation for analytics and innovation as they integrate with a suite of AWS analytics and artificial intelligence (AI) and machine learning (ML) services. Through these tools, insurers can utilize data lakes for a wide range of use cases, from traditional business reporting to advanced AI.

[Explore Analytics on AWS >](#)

A transformational element of this access to comprehensive data is the ability to build a 360-degree view of the customer, including their products and interactions across channels and devices. Customer information can be easily stored in a database, while a graph database, like **Amazon Neptune**, is ideal for managing the relationship model between customer, product, activity, agent, and desired outcome. A holistic view of the customer enables insurers to make significant advances in operational improvements and delivery of more modern, customer-centric experiences.

Innovation in action



Nationwide builds a governed data lake on AWS

OVERVIEW

Nationwide's on-premises data lake required costly and time-consuming infrastructure management and maintenance. The insurer also found that 80 percent of its stored data wasn't accessible.

SOLUTION

Nationwide built a governed data lake on AWS, with the aim of making all data findable, understandable, accessible, and secure via API-driven data catalogs.

IMPACT

Nationwide's governed data lake on AWS enabled it to build ML models to improve the speed and accuracy of insurance underwriting. The data lake also facilitated offering personalized tips for safe drivers. In addition, Nationwide was able to reduce storage costs.

[Learn more >](#)



PRIORITY 3

Increase digital engagement

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PRIORITY 3

Increase digital engagement

While insurance customers tend to have limited touchpoints with their carrier or broker beyond the purchase, renewal, and occasional claims or service transactions, insurers today have the opportunity to engage more frequently and proactively with customers at the point of need. Insurers that can successfully deliver omnichannel, hyper-personalized experiences at the right time through the right channel with the right message stand to gain immense benefits, including greater customer satisfaction and better retention. These experiences also drive improved operations and reduce underwriting and claims cycle times, which help to increase revenue and lower operating costs.

Personalized, digital experiences can improve customer satisfaction, retention, and net promoter scores

71%

Among customers 55 and older, 71% want to process claims through digital platforms, like voice or chat, instead of in person.³

50%

One negative experience is enough for 50% of customers to switch to a competitor.⁴

81%

Personalized services boost an insurer's retention rate to an incredible 81%.⁵

AWS supports your journey

AWS services and our community of AWS partners enable all insurance participants to provide customers a personalized experience.



Discovery

Target customers at the point of need

AWS and its partners can help insurers achieve a 360-degree view of customers that enables them to better understand behaviors and engage at the point of need. Insurers can build audience models by utilizing their zero and first-party customer data, partner with third-party data providers, and utilize services like **Amazon Clean Rooms** to deliver targeted marketing and ads that help customers select products.



Engagement

Build trust with proactive communication

With easier access to data and AWS ML tools—from **Amazon Connect Contact Lens** to **Amazon Personalize**—insurers can derive insights that enable them to meet customer expectations through personalized and timely experiences across transactions, such as upcoming payments or renewals. They can also strengthen relationships through contextually relevant offers and rewards, as well application or claims statuses and alerts. These capabilities can help to increase retention and drive organic growth through more targeted and successful cross-sell initiatives, all while driving down the traditional costs of engagement.



Digital Channels

Drive organic growth

AWS helps insurers to evolve their contact center and web portals into growth channels that enable cross-selling and the delivery of targeted offers, in the right channel at the right time. By using natural language processing (NLP) and sentiment analysis, as well as advanced ML modeling capabilities, insurers can develop more intelligent interactions with customers and turn service transactions into sales opportunities.

AWS supports your journey

AWS services and our community of AWS partners enable all insurance participants to provide customers a personalized experience.



Underwriting

Digitize risk selection

Insurers are reducing operating expenses and improving their risk selection by leveraging third-party data sources and AWS ML services to automate underwriting activities. For example, AWS enables insurers to accelerate the application intake process and reduce error and Not in Good Order (NIGO) numbers at scale with **Amazon Textract**, **Amazon Comprehend**, and **Amazon SageMaker**. Insurers also can build a deeper understanding of a risk profile during the underwriting process by leveraging AWS ML capabilities, as well as IoT sensor-based data. In addition, image and video recognition (computer vision) capabilities support the use of mobile devices to simplify customer interactions for documenting insured assets and providing detailed insight into coverables and risk profiles.

Claims Automation

Improve the customer experience and reduce costs

The claims process is a critical nexus where customer relationships are either strengthened or weakened. Claims automation in the cloud can afford immense benefits to both insurers and customers by reducing costs and ensuring a fast and frictionless claims process. AWS enables insurers to build an end-to-end automated claims solution, starting with receiving submissions from the customer's channel of choice using NLP and reducing fraud risk with ML services to automating adjudication and settlement with ML and **AWS Step Functions**.

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Innovation in action



Unum creates an omnichannel engagement platform on AWS

OVERVIEW

When Covid-19 hit, Unum needed to double down on its digital strategy to accommodate more digital interactions with customers.

SOLUTION

Unum began a journey to build an omnichannel customer engagement platform using AWS services, including **Amazon Connect**, **Amazon Pinpoint**, and **Amazon Lex**.

IMPACT

Early benefits include an increase in the use of self-service channels, improved economics of the contact center, and increased employee satisfaction with intuitive tools and simplified call center management.

[Learn more >](#)



When people are conversing with us using live chat, we are seeing 100 percent digital. It's not like they are chatting with us and then going back to calls or some other channel, so that has been awesome."

Gautam Roy

Senior Vice President and CTO, Unum



[Learn more](#)



PRIORITY 4

Re-imagine distribution

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PRIORITY 4

Re-imagine distribution

The sales journey across channels (direct-to-consumer, producer-led, affinity partners, etc.) remains a largely manual process, occasionally assisted by technology at isolated, inconsistent points. This creates significant challenges, including incomplete customer profiles and fragmented experiences. Many insurance products are also too complex for direct or digital distribution.

The shift in customer expectations, however, is driving insurers to re-imagine the distribution landscape. While insurers continue to mix and match between digital and traditional models, there is an increased focus on moving to direct channels and expanding the use of digitalized services. The end goal is to improve the customer experience by delivering a seamless and frictionless experience, whether through an insurer's own channels, third-party insurance distributors, or brand new channels outside of the industry.

Global embedded insurance gains momentum

\$722 Billion

Size of embedded insurance market—6x increase since 2021⁶

Build the foundation for a comprehensive platform

A comprehensive distribution solution needs to support all distribution channels



Direct-to-consumer

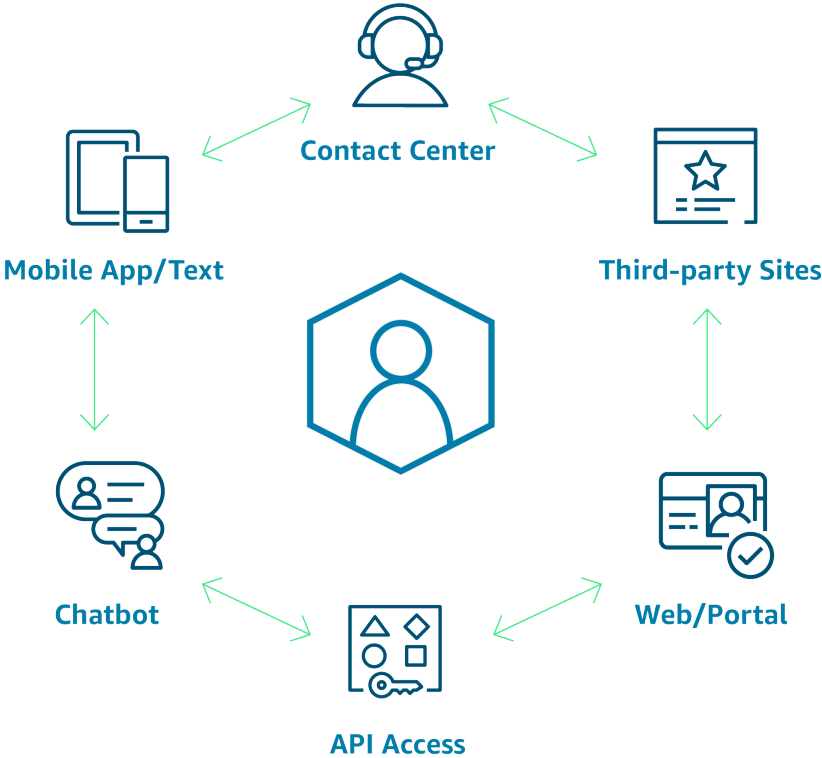


Producer-led

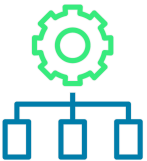


Third-party service providers

Customers need to be supported seamlessly across engagement channels



Each distribution solution needs an API ecosystem to connect with external partners



AWS supports your journey

Insurers can leverage AWS services, including AI and ML, to remove friction from the insurance buying process, from pre-sales to acceptance and issuance.

With AWS and our community of partners, insurers are building comprehensive distribution platforms that:

- Support all channels of distribution
- Support consumers seamlessly across engagement channels
- Allow insurers to connect with external partners easily, supporting direct digital channels, producer-led channels, and embedded insurance

Once the foundational platform is established, insurers can transform the sales journey to make it frictionless and low touch.

For example, AWS helps insurers simplify and personalize product selection and the application process by applying ML and NLP services to consolidate first- and third-party customer data. In addition, insurers can automate application processing and underwriting functions to reduce NIGO delays and fraud, while accelerating issuance with **Amazon Rekognition** for computer vision, **Amazon Textract** for text extraction, **Amazon Comprehend** for understanding unstructured data, and **AWS Step Functions** for workflow automation.



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Innovation in action



AXA builds its Insurance-as-a-Service platform on AWS

OVERVIEW

AXA Singapore created an Insurance-as-a-Service platform to make it easy for partner companies to offer insurance as end users are making related purchases, like booking travel, renting a car, or buying a house.

SOLUTION

The platform lets businesses integrate home, travel, and motor insurance products and services into their online platforms through an API call. The end user is then able to add on AXA insurance coverage in a few clicks.

IMPACT

AXA has forged partnerships to integrate its Insurance-as-a-Service platform into businesses, letting customers access insurance products when and how they want them.



The power of partners

AWS insurance customers can access a global partner community of independent software vendors (ISVs) with purpose-built solutions for the industry, including solutions sold through **AWS Marketplace**. **AWS Marketplace** offers a curated catalog of third-party solutions that can help insurers create innovative experiences with next-gen solutions, accelerate time to market and enhance the digital experience, comply with regulations, and unlock the potential of financial data.

[Explore Financial Services solutions in AWS Marketplace >](#)

AWS Data Exchange gives our financial services customers access to a wide variety of over 1,300 data sets and APIs relevant to the industry. This includes insurance claims; catastrophe data; personal income data; customer demographics; environment, social, and governance (ESG) ratings; and financial APIs.

[Explore third-party data solutions for Financial Services >](#)



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Explore the potential of generative AI

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PRIORITY 5

Explore the potential of generative AI

Generative artificial intelligence (AI) applications, which entered the public's imagination with the release of ChatGPT, are poised to accelerate innovation and transform business processes across the insurance industry and the financial services landscape at large.

Generative AI is a type of AI that can create new content and ideas, including human-like responses to natural language queries, reports and communications, document and content summaries, entity attributes, marketing copy, images, and videos. Like all AI, generative AI is powered by ML models—generative AI uses very large models, known as large language models (LLMs), that are pre-trained on vast amounts of data, commonly referred to as foundation models (FMs).

Insurance leaders are eager to explore and understand generative AI's potential for their business.

Potential generative AI use cases

- Improve internal and external insurance experiences
- Increase knowledge worker efficiency and automate processes
- Increase the value of unstructured content to drive insights and innovation

\$7 Trillion

Potential lift generative AI could drive in global GDP.⁷

AWS supports your journey

As insurance companies look to increase the speed, scalability, and flexibility of their products, processes, and systems, generative AI can help. Amazon and AWS have been working on AI and ML for more than 25 years and we are eager to share learnings with customers to improve their generative AI strategies and implementations. AWS has played a key role in democratizing ML and making it accessible to anyone who wants to use it, including more than 100,000 customers of all sizes and industries.

For insurers, AWS can help with:

Planning – Customizing the use of generative AI based on the insurer's unique use cases and needs. AWS works with each customer to choose and customize the right FM for each use case to ensure optimal performance and cost.

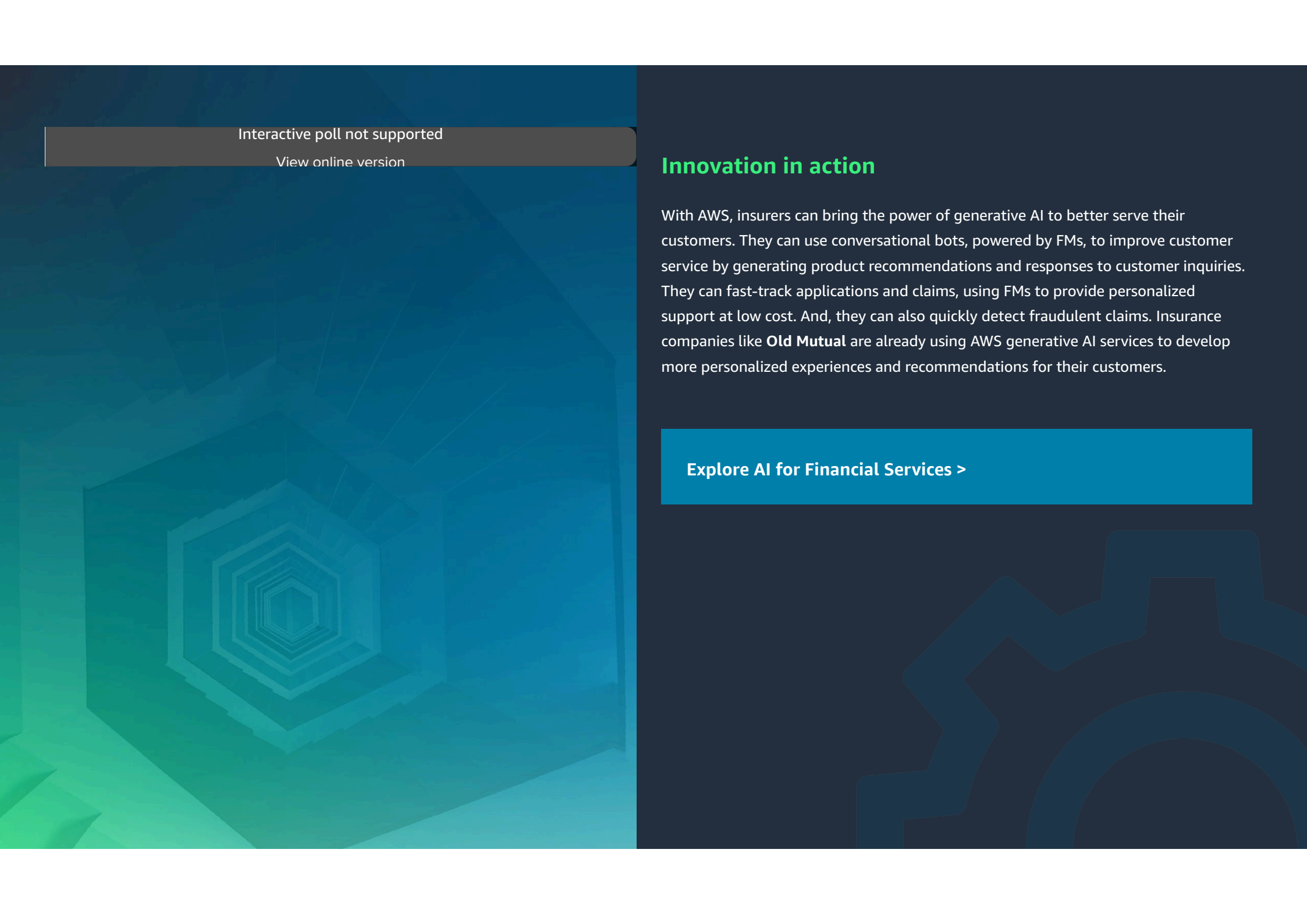
Tailoring generative AI applications – Giving insurers the flexibility to use multiple FMs through a single API and without having to manage the infrastructure in order to accelerate the development, testing, and deployment of AI applications. AWS offers insurers the flexibility to work with open-source models, including those from A121 Labs, Anthropic, Stability AI, Hugging Face, Cohere, and **Amazon Titan**, or build their own FMs.

Applying generative AI across all lines of business – AWS has the broadest and deepest portfolio of AI and ML services at all three layers of the stack, including infrastructure, tools, and purpose-built AI services.

Amazon and AWS have invested and innovated to offer the most performant, scalable infrastructure for cost-effective ML training and inference; developed **Amazon SageMaker**, which is the easiest way for all developers to build, train, and deploy models; and launched a wide range of services that allow customers to add AI capabilities, like image recognition, forecasting, and intelligent search to applications with a simple API call.

Maintaining data security and governance – Insurers can privately customize FMs using their organization's data and rely on established AWS services for model and data governance to maintain version control (MLOps).

Accelerating development – **Amazon CodeWhisperer** enables insurers to accelerate development by providing code recommendations in natural language, based on contextual information, such as prior code and comments.



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Innovation in action

With AWS, insurers can bring the power of generative AI to better serve their customers. They can use conversational bots, powered by FMs, to improve customer service by generating product recommendations and responses to customer inquiries. They can fast-track applications and claims, using FMs to provide personalized support at low cost. And, they can also quickly detect fraudulent claims. Insurance companies like **Old Mutual** are already using AWS generative AI services to develop more personalized experiences and recommendations for their customers.

[Explore AI for Financial Services >](#)

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As insurers tackle new and evolving business priorities, those that can make rapid progress in addressing them stand to gain considerable market advantages. Cloud technology is essential to achieving many of these priorities, giving insurers the agility, performance, scalability, and security to thrive today, build for the future, and leave a lasting legacy.

AWS cloud services and the solutions provided by our community of partners help insurers move confidently through their transformations to modernize core systems, harness data, and leverage AI and other emerging technology to create new ways of operating and delivering insurance. And, we empower insurers to address these priorities in managed and controlled increments, reducing risks and speeding time to business impact.

Explore AWS in Insurance >

About AWS for Financial Services

AWS is a pioneer at the intersection of financial services and technology, enabling our customers to optimize operations and accelerate innovation through the broadest set of services and partner solutions—all while adapting to ever-changing regulations. Thousands of financial services firms, from the fastest-growing fintechs to systemically important financial institutions, are redefining their future on AWS.

Sources

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Thank you for reading Insurance on the Cloud

AWS for Financial Services

Amazon Web Services (AWS) is a pioneer at the intersection of financial services and cloud technology, enabling our customers to optimize operations and accelerate growth through the broadest set of services and partner solutions—all while adapting to ever-changing regulations. Thousands of financial services firms—from the fastest-growing fintechs to systemically important financial institutions—are redefining their future on AWS. **For more information about AWS for Insurance, visit aws.amazon.com/financial-services/insurance**

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